Measuring Firm-Level Geopolitical Exposure

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Working paper. Comments welcome.

Abstract

Existing research has developed important measures of aggregate geopolitical risk, but it remains difficult to assess which individual firms are most exposed when that risk rises during wars, trade disputes, or other geopolitical shocks. I develop GeoX, a market-implied measure of firm-level geopolitical exposure based on firms' historical excess-return sensitivities to news-based geopolitical shock series over 2015–2023. Firms receive higher GeoX scores when their returns respond more strongly to geopolitical shocks. GeoX identifies economically interpretable exposure patterns, especially among oil, gas, and drilling firms. In out-of-sample validation, high-GeoX firms earn lower excess returns over the 2023–2025 period and experience larger negative cumulative abnormal returns after the October 2023 Middle East shock. The evidence suggests that firm-level geopolitical exposure is measurable, distinct from aggregate geopolitical risk, and informative about cross-sectional return behavior.

Keywords: Geopolitical risk; Political risk; Firm-level exposure; Stock returns; Asset pricing

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1 Introduction

Recent years have seen a series of major geopolitical shocks. The 2023–2024 Red Sea crisis disrupted global shipping lanes (Kamali et al., 2024), the October 7, 2023 Middle East shock was followed by broader regional conflict (Reuters, 2025), and the 2022 Russia–Ukraine conflict triggered sanctions and energy disruptions (International Energy Agency, 2024). These developments have increased the salience of geopolitical risk for market participants. For example, JPMorgan Chase CEO Jamie Dimon emphasized that the bank's "largest risk is geopolitical risk," and the bank later launched a Center for Geopolitics (Dimon, 2024; JPMorganChase, 2025); Citigroup CEO Jane Fraser described geopolitics as the "biggest shift in the world of finance" (Fraser, 2024). Survey evidence similarly shows that CEOs and executives rank geopolitics, trade policy, and industrial policy among the risks most likely to affect business operations (Dewar et al., 2023; Khatib et al., 2023; Nelson et al., 2025). These patterns raise the central measurement question of the paper: which firms are most exposed when geopolitical shocks occur?

Existing research has made substantial progress in measuring the evolution of geopolitical risk over time. Aggregate geopolitical-risk measures capture time-series variation in geopolitical risk (Caldara and Iacoviello, 2022), but they leave open the cross-sectional question of which firms are most exposed when that risk changes. This question is difficult because firm-level geopolitical exposure is latent: a firm's sensitivity to geopolitical shocks is not directly observable from aggregate indexes or standard accounting characteristics. Disclosure- and earnings-call-based measures move closer to the firm level by capturing what managers, firms, and analysts discuss (Hassan et al., 2019, 2024). Yet discussion is not the same as exposure, because it can also reflect issue salience, disclosure

incentives, and the broader information environment. This leaves a need for a firm-level measure that attempts to isolate underlying geopolitical exposure.

I address this gap by developing GeoX, a market-implied measure of firm-level geopolitical exposure. GeoX is market-implied because equity prices aggregate market participants' information and expectations about how geopolitical shocks affect firms' expected cash flows and required returns. When new geopolitical information arrives, firms with greater underlying exposure should experience larger return responses. GeoX uses these historical return sensitivities to estimate firm-level geopolitical exposure.

The construction proceeds in two steps. First, I use the *Global Database of Events, Language, and Tone* (Leetaru and Schrodt, 2013) to construct monthly country-level GeoRisk and GeoShock series for 14 geopolitically salient countries over the estimation period from 2015 to 2023. GeoRisk is a signed country-month score based on the share and tone of country-specific geopolitical news, and GeoShock is the monthly change in GeoRisk. Second, I estimate country-specific geopolitical betas that capture each firm's excess-return sensitivity to the standardized GeoShock series for each country. I then construct GeoX by aggregating the absolute values of statistically significant betas into a firm-level exposure score. Firms receive higher GeoX scores when their historical excess returns respond more strongly to geopolitical shocks associated with one or more country-level GeoShock series.

I evaluate GeoX in three complementary ways. First, I examine whether the measure produces economically interpretable cross-sectional rankings. The descriptive evidence points to clear ex-

posure patterns: energy-related firms dominate the highest-GeoX rankings, consistent with the cash-flow sensitivity of commodity-linked firms to geopolitical shocks (Cai et al., 2025; Caldara and Iacoviello, 2022; Palomba and Tedeschi, 2025). Second, I test whether pre-estimated GeoX scores are associated with firm excess returns outside the estimation period. During the out-of-sample period from September 2023 to June 2025, GeoX is associated with cross-sectional differences in excess returns. Third, I examine whether GeoX separates abnormal returns around a concentrated geopolitical event. In an event study around the October 7, 2023 Middle East shock, high-GeoX firms experience significantly larger negative cumulative abnormal returns than low-GeoX firms.

This paper makes three contributions. First, it adds a cross-sectional dimension to the geopolitical risk literature. Existing aggregate measures identify when geopolitical risk is elevated, but they do not directly speak to which firms are most exposed when geopolitical conditions change. GeoX makes this cross-sectional variation measurable by estimating firm-level exposure to country-level geopolitical shocks. Second, GeoX offers a market-implied complement to the political risk and disclosure literature. Disclosure-based measures capture what firms, managers, and analysts discuss, which may reflect not only underlying exposure but also issue salience, disclosure incentives, and the broader information environment. Because GeoX is inferred from return responses rather than communication, it provides a benchmark for studying when disclosed or discussed risk aligns with, or diverges from, market-implied exposure. Third, the empirical validation suggests that firm-level geopolitical exposure has return-relevant content. High-GeoX firms earn lower excess returns during a geopolitically turbulent out-of-sample period and respond more strongly to a concentrated geopolitical shock. These findings suggest that geopolitical exposure is a measurable firm attribute with implications for asset pricing, disclosure, and corporate risk analysis.

The rest of the paper proceeds as follows. Section 2 develops the conceptual framework, distinguishes aggregate geopolitical risk from firm-level exposure, reviews existing measures and the measurement gap, and discusses economic sources of firm-level exposure. Section 3 describes the construction of country-level GeoRisk and GeoShock series and the estimation of the firm-level GeoX measure. Section 4 evaluates GeoX using descriptive rankings, out-of-sample return regressions, and an event study. Section 5 concludes.

2 Conceptual Framework and Related Literature

This section develops the conceptual distinction between aggregate geopolitical risk and firm-level geopolitical exposure. It first defines the two concepts, then explains why existing measures do not directly identify exposure, and finally discusses the economic channels through which exposure differs across firms.

2.1 Aggregate Geopolitical Risk and Firm-Level Geopolitical Exposure

Following Caldara and Iacoviello (2022), I treat geopolitical risk as encompassing the threat, realization, and escalation of adverse events involving wars, terrorism, and tensions among states or political actors that disrupt the peaceful course of international relations. This definition motivates two concepts that are central to the paper. Aggregate geopolitical risk captures time-series variation in geopolitical conditions. Firm-level geopolitical exposure captures a firm's sensitivity to geopolitical shocks. Conceptually, firm-level geopolitical risk depends on two components: aggregate

geopolitical risk and firm-level geopolitical exposure:

$$\text{Firm-Level Geopolitical Risk}_{i,t} \propto \text{Aggregate Geopolitical Risk}_t \times \text{Firm-Level Geopolitical Exposure}_i.$$

This decomposition is a conceptual device, not a structural asset-pricing equation. Its purpose is to separate a time-series component, aggregate geopolitical risk, from a cross-sectional component, firm-level geopolitical exposure. Exposure can remain high even when aggregate geopolitical risk is low, and an increase in aggregate geopolitical risk can have limited return implications for firms with low exposure.

2.2 Existing Measures and the Measurement Gap

Existing measures related to geopolitical risk differ in the informational object they capture. Some measure aggregate geopolitical conditions, others capture broader uncertainty or volatility, and others use firm communication to measure discussion and perceived risk.

The first group measures aggregate geopolitical conditions. Caldara and Iacoviello (2022) develop the Geopolitical Risk (GPR) index by measuring the share of newspaper articles that jointly mention geopolitical and risk-related terms. Sheng et al. (2025) extend this approach by incorporating trade-war language, and GEOVOL captures geopolitical volatility (Engle and Campos-Martins, 2020). These measures identify periods when geopolitical risk is elevated or geopolitical conditions become more volatile, making them useful for studying time-series variation in geopolitical risk and its market-wide effects.

Broader uncertainty and volatility measures provide related but distinct information. The Economic Policy Uncertainty index (Baker et al., 2016), the World Uncertainty Index (Ahir et al., 2022), and the News Implied Volatility Index (Manela and Moreira, 2017) capture policy uncertainty, global uncertainty, or news-implied volatility. These measures matter because geopolitical shocks often coincide with broader uncertainty and volatility, but their object is broader than geopolitical risk specifically. Taken together, aggregate geopolitical-risk indices and broader uncertainty measures are informative about the aggregate environment, but they do not directly isolate underlying firm-level exposure when geopolitical conditions change.

A third approach uses earnings-call communication to measure discussed or perceived risk. Zhou (2025) uses earnings calls to measure firm-level geopolitical risk, Hassan et al. (2019) use earnings-call discussions to measure firm-level political risk, and Hassan et al. (2024) use earnings conference calls to measure perceived country risk. The mapping from communication to exposure, however, is indirect because geopolitical and political conditions can affect the communication environment itself. Geopolitical tensions can alter investors' access to conference-call information (Lu et al., 2025), while political uncertainty can shape firms' voluntary disclosure behavior (Bird et al., 2023). As a result, earnings-call discussion reflects not only firms' underlying exposure, but also variation in information access and disclosure incentives.

These approaches leave a distinct measurement problem: latent firm-level geopolitical exposure. Returns provide a complementary signal because market prices aggregate dispersed information about the firm-specific exposure channels through which geopolitical shocks affect expected cash

flows and required returns. If those channels differ across firms, the same shock should generate different excess-return sensitivities. GeoX uses these differential sensitivities to infer firm-level geopolitical exposure.

2.3 Economic Sources of Firm-Level Geopolitical Exposure

Firm-level geopolitical exposure arises because the same geopolitical shock can have different implications for firms' expected cash flows and required returns. These differences should appear in returns when markets incorporate geopolitical information into prices. Return sensitivities therefore provide a market-implied signal of exposure. Rare-disaster and political-uncertainty models provide the theoretical basis for this pricing logic, showing how low-probability adverse events and political uncertainty can affect prices before losses are realized (Rietz, 1988; Barro, 2006; Liu et al., 2005; Pastor and Veronesi, 2013; Gulen and Ion, 2016). Empirical evidence points in the same direction: political crises and war-related news are associated with lower prices, higher volatility, and higher expected returns (Berkman et al., 2011; Hirshleifer et al., 2025; Federle et al., 2026; Carney et al., 2024). The remaining question is why firms differ in their return sensitivity to the same geopolitical shock.

Firm-level exposure can arise through several related economic links. Production networks can propagate shocks across firms and sectors, especially when suppliers provide specific inputs that are difficult to replace (Acemoglu et al., 2012, 2016; Barrot and Sauvagnat, 2016). Evidence from the Great East Japan Earthquake shows that disruptions travel through supply chains rather than remaining localized to the initially affected firms or regions (Carvalho et al., 2021). These links

can also matter for asset prices: global production networks transmit international stock-return spillovers in other settings (di Giovanni and Hale, 2022), firms exposed to China-related trade experienced larger market-value declines during the U.S.–China trade war (Huang et al., 2023), and export controls generated negative equity-market reactions and real costs for affected U.S. suppliers (Crosignani et al., 2026). Direct country-specific ties matter as well. Firms with greater bilateral exposure experienced larger stock-price declines when Sino-Japanese relations deteriorated (Fisman et al., 2014), and geopolitical volatility in bilateral political relations is associated with lower subsidiary investment, employment, and local sales in host countries (Adarkwah et al., 2024). More broadly, geopolitical shocks can affect firm behavior through institutional, regulatory, legal, and governance mechanisms (Andrews et al., 2026).

Taken together, these mechanisms imply that firm-level geopolitical exposure is economically meaningful but only partially captured by accounting data, industry classifications, or disclosures. Many relevant firm-country ties are private, dynamic, or only partially observable. A market-implied approach therefore provides a practical way to infer exposure from how firms' returns respond to geopolitical shocks.

3 Measuring Firm-Level Geopolitical Exposure

This section describes the construction of GeoX. The measure is built in two stages. The first stage constructs monthly country-level GeoRisk and GeoShock series from geopolitical news coverage, where GeoRisk captures the level of country-specific geopolitical conditions and GeoShock captures monthly changes in those conditions. The second stage estimates firms' excess-return sensitivities

to those country-level geopolitical shocks and uses the resulting country-specific geopolitical betas to construct a firm-level exposure measure. Throughout the section, the objective is to isolate firm-level geopolitical exposure rather than aggregate geopolitical risk itself.

3.1 Country-Level GeoRisk and GeoShock Construction

The first stage constructs country-level GeoRisk and GeoShock series from international geopolitical news coverage. I use the *Global Database of Events, Language, and Tone* (GDELT) (Leetaru and Schrodt, 2013), a large-scale news database that monitors global media coverage in near real time. The key input is GDELT's Global Knowledge Graph 2.0 (GKG 2.0), a structured metadata layer that extracts entities, themes, locations, publication dates, and sentiment from unstructured news articles. This structure makes it possible to construct separate country-level GeoRisk and GeoShock series rather than relying on a single aggregate geopolitical-risk index. Prior research has used GDELT to study political protest activity and geopolitical tensions, including protest activity during Egypt's Arab Spring (Acemoglu et al., 2017) and the effect of geopolitical tensions on foreign investors' access to firm information (Lu et al., 2025).

The construction of the country-level series proceeds in four steps. First, I extract publication dates, country references, themes, and tone measures from each GKG record. From the full universe of GKG themes, I select a curated set of geopolitical themes spanning military, political, economic, informational, social, and resource-related domains (see Appendix A).

Second, I apply a proximity filter to reduce incidental co-mentions. A country reference and a

geopolitical theme must appear within 500 characters of one another in the article text. An article is therefore classified as a geopolitical article for country c only when this condition is satisfied.

Third, I aggregate articles to the country-month level using publication dates. For each country-month pair, I compute both the number of geopolitical articles and the total number of articles mentioning the country. Their ratio defines the monthly geopolitical news share:

$$\text{GeoShare}_{c,t} = \frac{|\text{GeoArticles}_{c,t}|}{|\text{TotalArticles}_{c,t}|}$$

I also compute the average tone of geopolitical articles within the country-month:

$$\text{GeoTone}_{c,t} = \frac{1}{|\text{GeoArticles}_{c,t}|} \sum_{\text{article} \in \text{GeoArticles}_{c,t}} \text{Tone}_{\text{article}}$$

Fourth, I define the country-level GeoRisk measure as the product of geopolitical news share and average geopolitical tone:

$$\text{GeoRisk}_{c,t} = \text{GeoShare}_{c,t} \times \text{GeoTone}_{c,t}$$

GeoRisk is therefore a signed measure of the level of country-specific geopolitical conditions in month t. Because GeoRisk incorporates signed article tone, more negative values indicate greater geopolitical risk, while more positive values reflect relatively cooperative or de-escalatory geopolitical conditions. Values near zero indicate either limited geopolitical news coverage or relatively neutral tone.

I define the country-level geopolitical shock, $\text{GeoShock}_{c,t}$, as the monthly change in GeoRisk:

$$\text{GeoShock}_{c,t} = \text{GeoRisk}_{c,t} - \text{GeoRisk}_{c,t-1}$$

GeoShock captures month-to-month changes in GeoRisk. Under this construction, adverse geopolitical developments are reflected in GeoRisk becoming more negative, while easing or cooperative developments are reflected in it becoming more positive. To ensure comparability across countries with different levels of media attention and tone dispersion, I standardize each country's GeoRisk and GeoShock series into z-scores over the estimation period. Figures 1–14 present the resulting standardized series.

I apply this procedure to a curated set of 14 countries that are major sources of geopolitical variation, denoted by $\mathcal{C}$, over the estimation period from February 19, 2015, when GDELT GKG 2.0 first became available, through August 31, 2023. The set includes the five permanent members of the United Nations Security Council: the United States, China, Russia, the United Kingdom, and France (United Nations Security Council, 2025). It also includes major regional powers with substantial economic or military influence: Japan, Germany, India, Brazil, Saudi Arabia, Israel, and South Korea (Tian et al., 2024; World Bank, 2025). Finally, it includes North Korea and Iran, two heavily sanctioned countries that are frequent subjects of geopolitical news (U.S. Department of the Treasury, Office of Foreign Assets Control, 2025b,a).

The country set is intentionally selective rather than exhaustive. The goal is not to include every internationally relevant country, but to capture major sources of geopolitical variation that are

likely to generate global spillovers and affect multinational firms. Other advanced economies, such as Australia, Canada, Italy, and Spain, matter for global markets, but their geopolitical role is often mediated through alliances and partnerships.

I conduct two diagnostic checks to assess whether the country-level series capture meaningful geopolitical variation. First, I manually inspect large movements in each country's GeoRisk series. For each country, I identify months in which the absolute standardized GeoRisk score exceeds two and compare those months with contemporaneous geopolitical developments reported in the news. Large deviations generally correspond to identifiable geopolitical developments, mitigating the concern that the series are driven primarily by media noise or coverage imbalances.

Second, I examine pairwise correlations across countries to assess whether the series capture distinct sources of geopolitical variation rather than a single global factor. Figure 15 shows that correlations are generally low to moderate for both the GeoRisk series (Panel A) and the GeoShock series (Panel B), with no evidence of severe collinearity. These patterns support the use of separate country-level GeoShock series in the firm-level exposure estimation.

3.2 Firm-Level Exposure Estimation

The second stage estimates firm-level geopolitical exposure using the country-level GeoShock series constructed in the previous subsection. The sample consists of firms that were constituents of the S&P 500 at any point during the estimation period, including both entrants and exits. Requiring at least 48 months of return observations yields a final sample of 621 firms. Table 1 reports summary

statistics for the estimation sample.

For each firm i, I estimate the following time-series regression of monthly excess returns on the full set of standardized country-level GeoShock series:

$$R^e_{i,t} = \alpha_i + \sum_{c \in \mathcal{C}} \beta_{i,c} \, \text{GeoShock}_{c,t} + \varepsilon_{i,t}.$$

In this specification, $R^e_{i,t}$ denotes the monthly return of firm i in excess of the risk-free rate in month t, α_i is a firm-specific intercept, $\text{GeoShock}_{c,t}$ is the standardized geopolitical shock for country c in month t, and $\varepsilon_{i,t}$ is the error term. I use monthly returns to match the frequency of the GeoShock series and express them in excess of the risk-free rate to remove the common risk-free component from firm returns. The coefficient $\beta_{i,c}$ is the firm's country-specific geopolitical beta. It measures how firm i's excess returns move with geopolitical shocks associated with country c, conditional on the other GeoShock series included in the regression. The sign captures the direction of the response, while the magnitude captures the strength of the firm's return sensitivity to geopolitical shocks associated with that country.

The regression specification is intentionally sparse because the objective is not to maximize return-model fit, but to estimate a reduced-form measure of firms' geopolitical exposure. Including controls such as oil prices, sector returns, or standard asset-pricing factors could absorb part of the geopolitical transmission mechanism because those variables may themselves respond to geopolitical shocks. In this respect, the design parallels the estimation of a stock's market beta.

After estimating the country-specific geopolitical betas, I aggregate them into a single firm-level exposure score:

$$\text{GeoX}_i = \sum_{c \in \mathcal{C}} \left| \hat{\beta}_{i,c} \right| \times \mathbf{1}\{p_{i,c} < 0.10\}.$$

Here, $\hat{\beta}_{i,c}$ is the estimated country-specific geopolitical beta for firm i, and $p_{i,c}$ is the associated regression p-value. GeoX sums the absolute values of country-specific geopolitical betas that are statistically significant at the 10% level. Taking absolute values captures exposure magnitude regardless of whether geopolitical shocks associated with a given country increase or decrease the firm's returns, because the objective is to measure the extent of firms' return responses to geopolitical shocks rather than the direction of those responses. The significance threshold serves as a screening device that reduces the influence of weakly estimated sensitivities.

This aggregation produces a transparent firm-level exposure measure without imposing ex ante country weights. Country importance instead enters through the magnitude and statistical significance of the estimated geopolitical betas. The resulting GeoX scores form the basis of the descriptive and empirical validation analyses in the next section.

4 Empirical Validation of GeoX

This section evaluates whether GeoX has the empirical properties expected of a firm-level geopolitical exposure measure. The validation proceeds in three steps. First, I examine whether GeoX identifies economically interpretable cross-sectional exposure patterns. Second, I test whether GeoX is associated with cross-sectional differences in firm excess returns during an out-of-sample period

characterized by elevated geopolitical tensions. Third, I conduct an event study around the October 7, 2023 Middle East shock to examine whether firms with greater measured exposure exhibit stronger abnormal-return responses. Together, these exercises assess the descriptive, out-of-sample, and event-study evidence on GeoX.

4.1 Descriptive Evidence

Table 2 reports the 25 firms with the highest GeoX scores. The top-ranked firms are concentrated in energy-related industries, especially oil, gas, and drilling. This pattern is economically plausible because commodity-linked firms are sensitive to geopolitical disruptions through expected cash flows and commodity-price movements. At the same time, individual-firm rankings may partly reflect firm-specific volatility or estimation noise unrelated to persistent geopolitical exposure. I therefore treat the top-firm list as suggestive and place more weight on industry-level patterns that aggregate across firms.

The industry-level evidence supports the same interpretation. Figure 16 shows that drilling oil and gas wells and crude petroleum and natural gas have median GeoX scores of 14.6 and 7.6, respectively, far above other industries. At the broader SIC division level, Figure 17 shows that mining is the most exposed division, followed by construction and retail trade. Although descriptive evidence is not a formal validation test, these patterns provide an economically intuitive cross-sectional benchmark: the industries identified by GeoX align with plausible channels of geopolitical exposure.

4.2 Out-of-Sample Regression Evidence

The second validation exercise examines whether GeoX is associated with cross-sectional variation in firm excess returns beyond the estimation period used to construct it. I therefore study the out-of-sample period from September 2023 to June 2025. This period includes several major geopolitical developments that were not part of the estimation period, including the October 7, 2023 Middle East shock, the subsequent regional conflict, the 2023–2024 Red Sea crisis, continued instability associated with the Russia–Ukraine conflict, and the escalation of U.S. trade tariffs in early 2025. If GeoX captures economically meaningful firm-level geopolitical exposure, firms with higher measured exposure should exhibit different excess-return behavior during this period.

I estimate a pooled OLS regression of firm-level excess returns on GeoX, month fixed effects, and interactions between GeoX and the standardized country-level GeoShock series:

$$R^e_{i,t} = \alpha + \gamma_t + \beta\,\mathrm{GeoX}_i + \sum_{c\in\mathcal{C}} \delta_c\,(\mathrm{GeoX}_i \times \mathrm{GeoShock}_{c,t}) + \varepsilon_{i,t}$$

where $R^e_{i,t}$ denotes the monthly return of firm i in excess of the risk-free rate in month t, γ_t are month fixed effects that absorb market-wide shocks common to all firms, GeoX_i is the firm's GeoX score estimated over the estimation period and held fixed during the out-of-sample period, and $\mathrm{GeoShock}_{c,t}$ is the standardized country-level GeoShock series for country c in month t.

GeoX is held fixed at its estimation-period value throughout the out-of-sample period, so the re-

sults are not driven by re-estimation during the validation window. Month fixed effects absorb shocks common to all firms, including the standalone country-level GeoShock series, so identification comes from differences across firms within the same month. The coefficient β captures the baseline excess-return difference associated with GeoX after controlling for month-level common shocks, while δ_c captures whether high-GeoX firms respond differently to changes in the GeoShock series for country c. Because GeoShock is signed, the sign of δ_c determines whether high-GeoX firms perform relatively better or worse as the country-level GeoShock series changes. A positive δ_c implies that high-GeoX firms perform relatively better when the GeoShock series improves and relatively worse when it deteriorates; a negative δ_c implies the opposite.

Table 3 reports the results. The baseline coefficient on GeoX is negative and marginally significant at the 10% level ($\beta = -0.163$, $p = 0.084$), suggesting that firms with higher exposure earned lower excess returns than firms with lower exposure during the out-of-sample period, after controlling for month-level common shocks. This finding is consistent with high-exposure firms underperforming during a period characterized by elevated geopolitical risk and repeated geopolitical shocks.

The interaction terms reveal additional cross-country heterogeneity in how high-exposure firms respond to geopolitical shocks. The interaction with the Saudi Arabia GeoShock series is negative and statistically significant ($\delta_{SA} = -0.903$, $p = 0.032$), implying that high-GeoX firms tend to perform relatively better when the Saudi Arabia GeoShock series deteriorates. This pattern may reflect Saudi Arabia's central role in global energy markets, where adverse geopolitical developments can raise oil prices and improve expected cash flows for energy-related firms with greater geopolitical exposure. In contrast, the interaction with the Israel GeoShock series is positive and statistically

significant ($\delta_{IS} = 0.453$, $p = 0.026$), implying that high-GeoX firms tend to underperform when the Israel GeoShock series deteriorates. This pattern may reflect broader regional instability and heightened geopolitical uncertainty affecting firms with greater measured exposure. The interaction with the China GeoShock series is also positive and significant at the 10% level, while the Germany interaction is positive but not statistically significant at conventional levels. Overall, the out-of-sample regression evidence suggests that pre-estimated GeoX scores contain information about cross-sectional excess returns during the validation window.

4.3 Event-Study Evidence

The third validation exercise is an event study around the October 7, 2023 Middle East shock. This setting provides a concentrated test of whether firms with greater measured geopolitical exposure exhibit stronger return responses following a major geopolitical event. The shock is particularly informative because it was a globally salient geopolitical event that occurred outside the GeoX estimation period. If GeoX measures cross-sectional exposure, firms with higher GeoX scores should exhibit larger abnormal return responses following the shock.

The event study compares cumulative average abnormal returns (CAARs) for firms in the top and bottom quartiles of nonzero GeoX over the 30 trading days following the shock. Abnormal returns are residuals from the Fama–French three-factor model augmented with momentum, estimated over the 250 trading days preceding the event window. I require at least 200 valid daily return observations and impose a 20-trading-day gap before the event window to reduce contamination from information leakage.

Figure 18 presents the results. Following the October 7 Middle East shock, the CAAR trajectories diverge sharply: low-GeoX firms earn modestly positive abnormal returns, while high-GeoX firms experience persistent losses reaching nearly -5% after 30 trading days. The difference in CAARs becomes significantly negative within the first two weeks and remains negative throughout the event window, with the gap approaching -6% at its trough. This event-study separation is consistent with the exposure ranking identifying firms that are more sensitive to the shock.

Taken together, the three exercises show that GeoX produces interpretable cross-sectional rankings, contains out-of-sample return information, and separates firms' abnormal returns around a concentrated geopolitical shock.

5 Conclusion

This paper develops GeoX, a market-implied measure of firm-level geopolitical exposure. The central distinction throughout the paper is between aggregate geopolitical risk, which varies over time, and firm-level geopolitical exposure, which varies across firms. Existing aggregate measures identify when geopolitical risk is elevated, while disclosure-based measures capture discussion and perceived salience of geopolitical issues. GeoX addresses this measurement gap by using firms' historical excess-return sensitivities to country-level geopolitical shocks to infer firm-level geopolitical exposure.

The construction proceeds in two stages. First, I use GDELT news data to construct country-level GeoRisk and GeoShock series for 14 countries that are major sources of geopolitical variation. GeoRisk captures country-level geopolitical conditions, while GeoShock captures changes in those conditions. Second, I estimate firms' historical excess-return sensitivities to the standardized GeoShock series and aggregate statistically significant country-specific geopolitical betas into a firm-level exposure score. Firms receive higher GeoX scores when their returns historically respond more strongly to geopolitical shocks associated with the countries to which they are most sensitive.

The empirical validation exercises provide three main findings. First, GeoX identifies economically interpretable cross-sectional exposure patterns. Firms and industries with high GeoX scores are concentrated in sectors with plausible geopolitical exposure channels, especially oil, gas, and drilling industries whose expected cash flows are sensitive to geopolitical disruptions and commodity-price movements. Second, GeoX is associated with cross-sectional differences in out-of-sample firm excess returns during the 2023–2025 period, which includes several major geopolitical developments outside the estimation window. Third, in the event study around the October 7, 2023 Middle East shock, firms with higher GeoX scores experience significantly more negative cumulative abnormal returns than firms with lower GeoX scores. Taken together, these findings suggest that GeoX is interpretable across firms and informative about return behavior outside the estimation period, while leaving open the precise cash-flow and required-return channels.

This paper makes three contributions. First, it adds a cross-sectional dimension to the geopolitical risk literature. Existing aggregate measures identify when geopolitical risk is elevated, but the evidence here shows that firms differ meaningfully in their exposure within the same aggregate

geopolitical environment. Studying geopolitical risk only as a time-series phenomenon therefore misses important heterogeneity: two firms can face the same geopolitical shock and respond very differently. Second, GeoX offers a market-implied complement to disclosure-based measures of political and geopolitical risk. Because GeoX is inferred from return responses rather than communication, it provides a benchmark for studying whether what firms, managers, and analysts discuss aligns with the exposure implied by market prices. Third, the out-of-sample return evidence and event study suggest that firm-level geopolitical exposure has return-relevant content. Whether this reflects compensation for risk, mispricing, or both remains open for future research, but the evidence suggests that geopolitical exposure is a measurable firm attribute with implications for asset pricing, disclosure, and corporate risk analysis.

GeoX also has practical implications for investors, managers, and policymakers. Investors and risk managers can use the measure to assess how geopolitical exposure is distributed across firms, portfolios, and industries. Managers can compare market-implied exposure with internal assessments of geopolitical vulnerability and supply-chain risk. For regulators and policymakers, the cross-sectional distribution of GeoX may help identify where geopolitical exposure is concentrated across firms and sectors.

Several limitations remain. First, GeoX is estimated using historical S&P 500 constituents, so its external validity for smaller firms, private firms, and non-U.S. markets remains to be tested. Second, the country-level GeoRisk and GeoShock series depend on geopolitical news coverage in GDELT, which may contain coverage imbalances and media noise despite the diagnostic checks. Third, GeoX is a reduced-form exposure measure: it captures market-implied sensitivity to geopo-

litical shocks, but it does not separately identify the economic channels through which exposure operates. Future research can extend the framework to other markets, refine the construction of country-level geopolitical shocks, and study how firms disclose, manage, and mitigate geopolitical exposure, and how markets price it under changing geopolitical conditions.

Overall, the evidence suggests that geopolitical exposure is measurable at the firm level using market responses to geopolitical shocks. Treating geopolitical risk only as an aggregate time-series phenomenon misses important cross-sectional variation in how firms respond when geopolitical shocks occur. GeoX provides a market-implied framework for measuring that variation and studying how firms differ in their exposure to geopolitical shocks.

6 Figures and Tables

Notes for Figures 1–14: Each figure plots the standardized monthly country-level GeoRisk series in Panel A and the standardized monthly GeoShock series in Panel B for the indicated country over 2015–2023. GeoRisk is a signed country-month measure based on the share and tone of country-specific geopolitical news coverage; lower values indicate more adverse geopolitical conditions. GeoShock is the monthly change in GeoRisk. Both series are standardized within country over the estimation period.

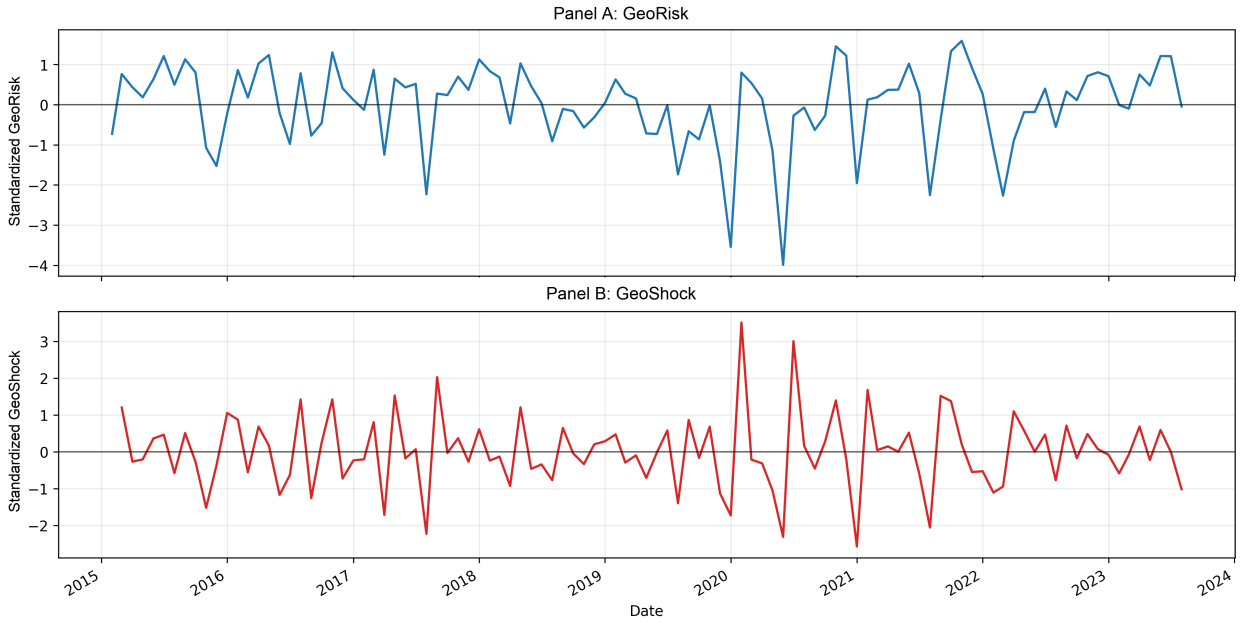


Figure 1: Country-Level GeoRisk and GeoShock Series for the United States

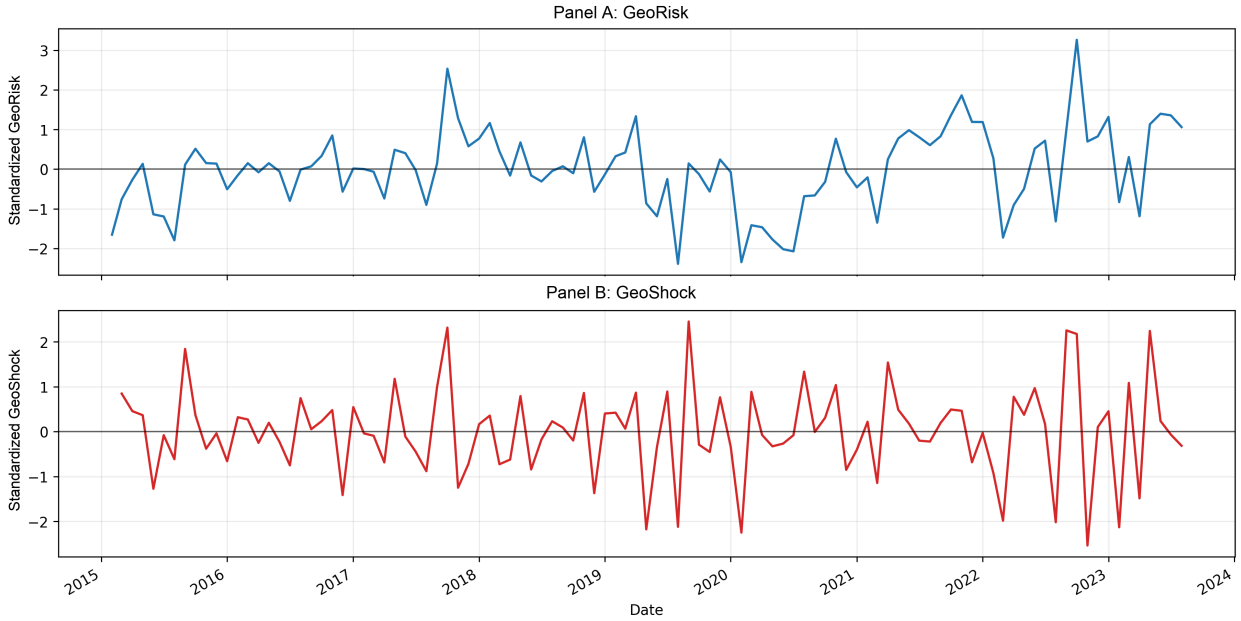


Figure 2: Country-Level GeoRisk and GeoShock Series for China

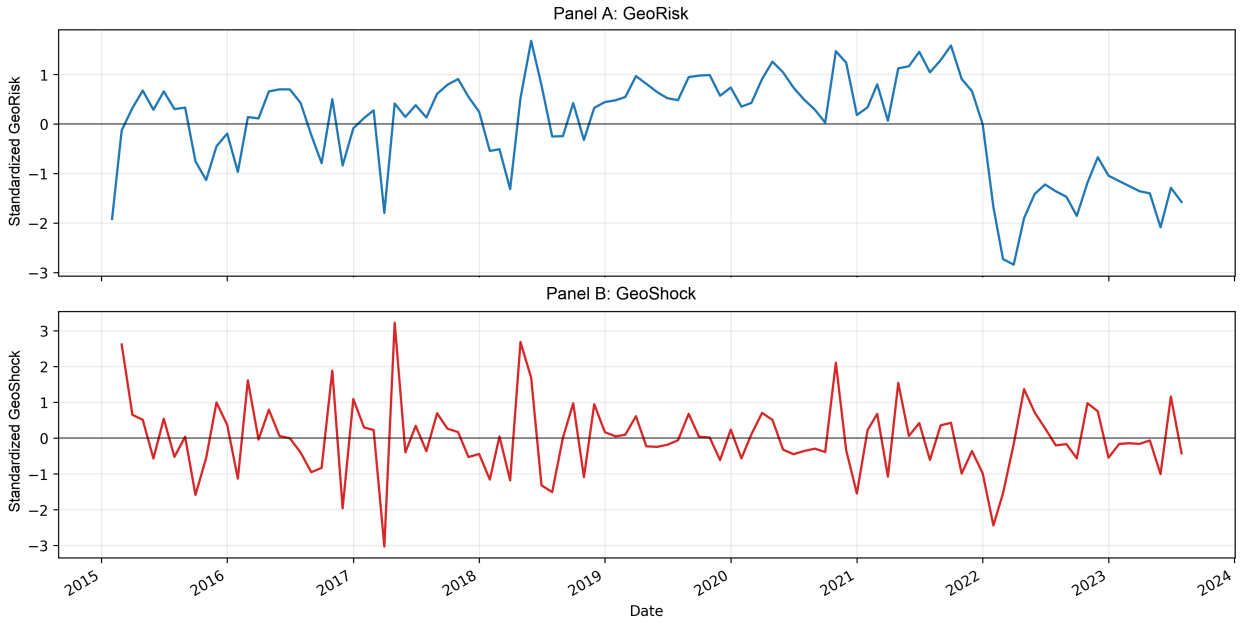


Figure 3: Country-Level GeoRisk and GeoShock Series for Russia

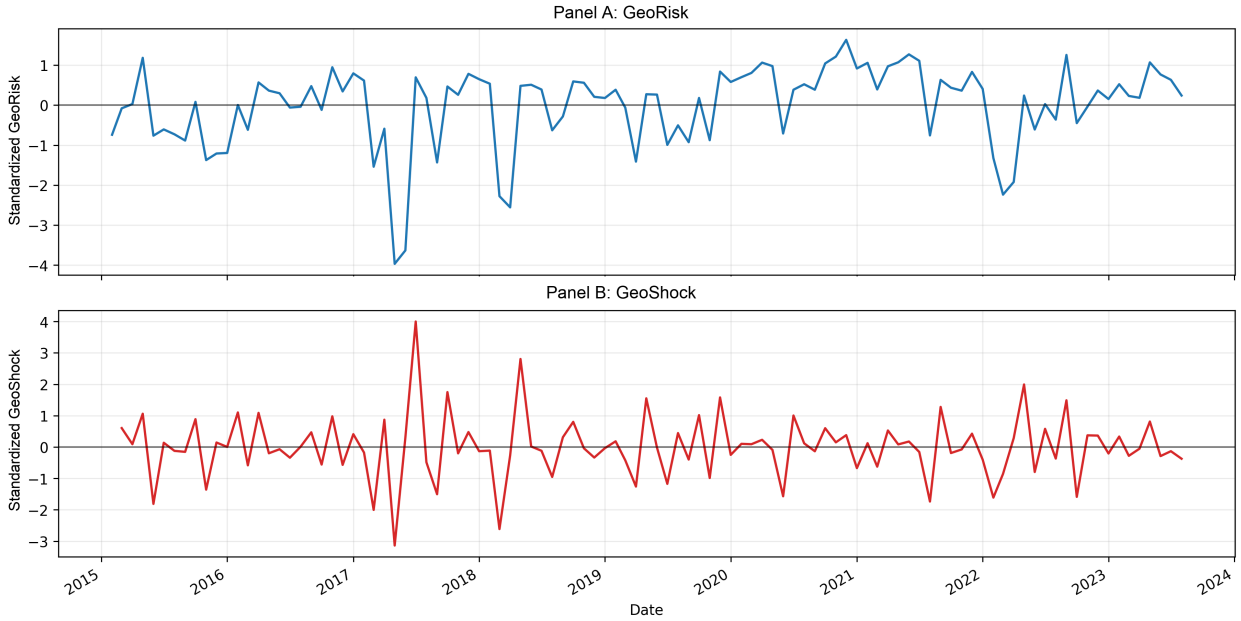


Figure 4: Country-Level GeoRisk and GeoShock Series for the United Kingdom

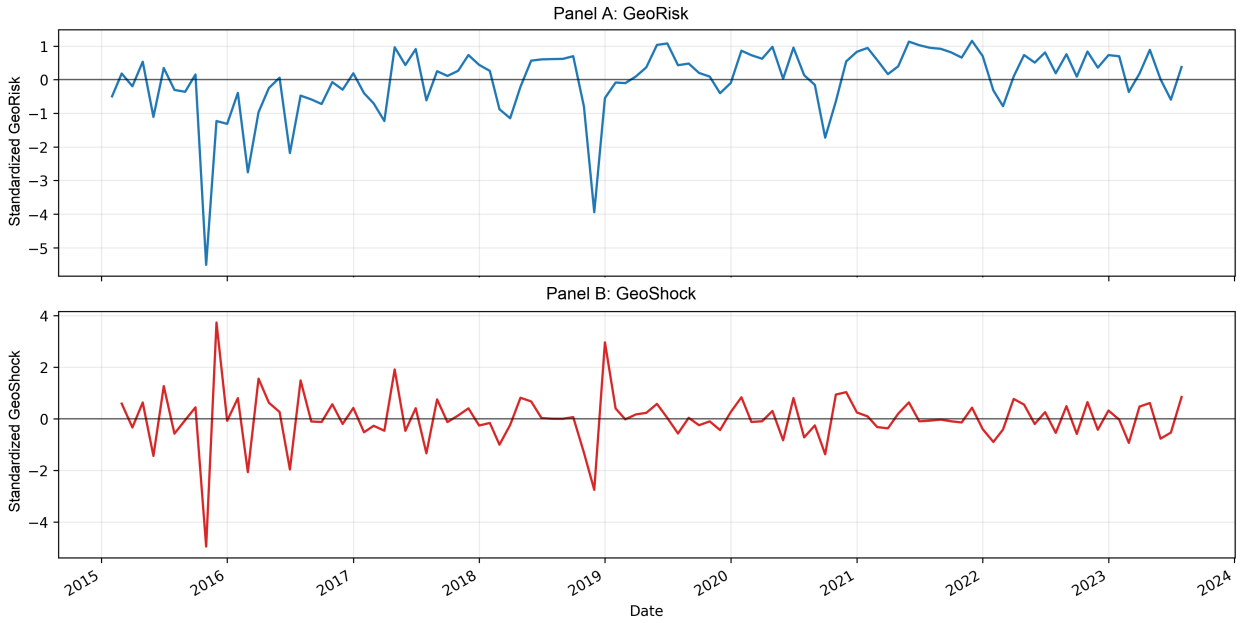


Figure 5: Country-Level GeoRisk and GeoShock Series for France

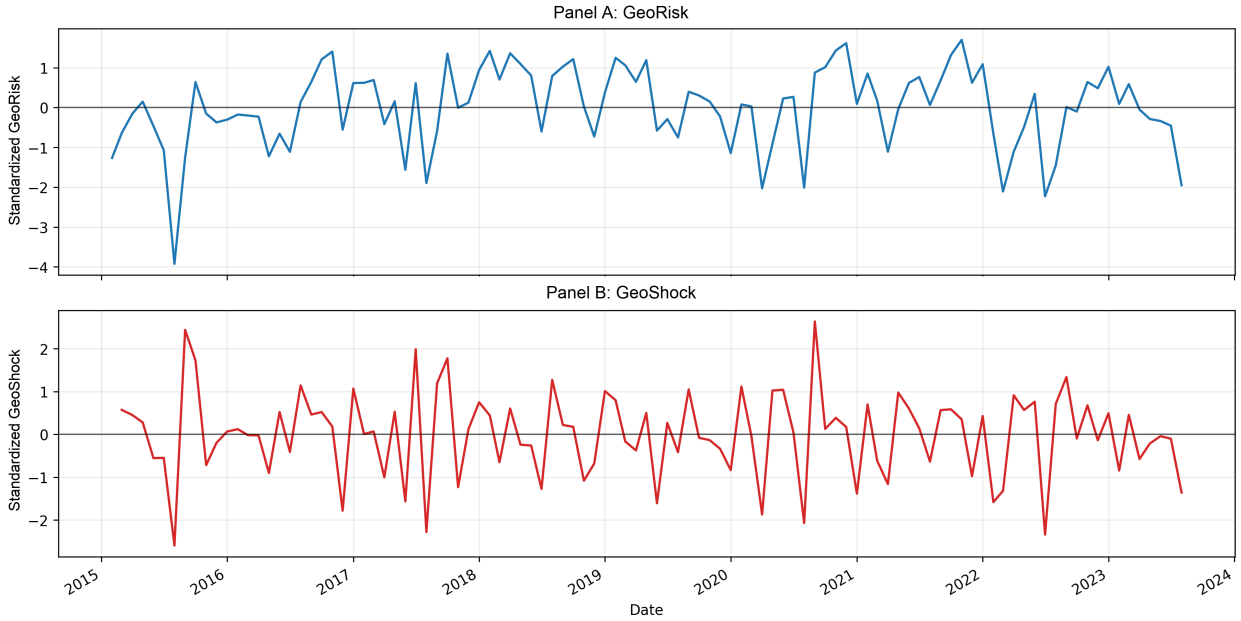


Figure 6: Country-Level GeoRisk and GeoShock Series for Japan

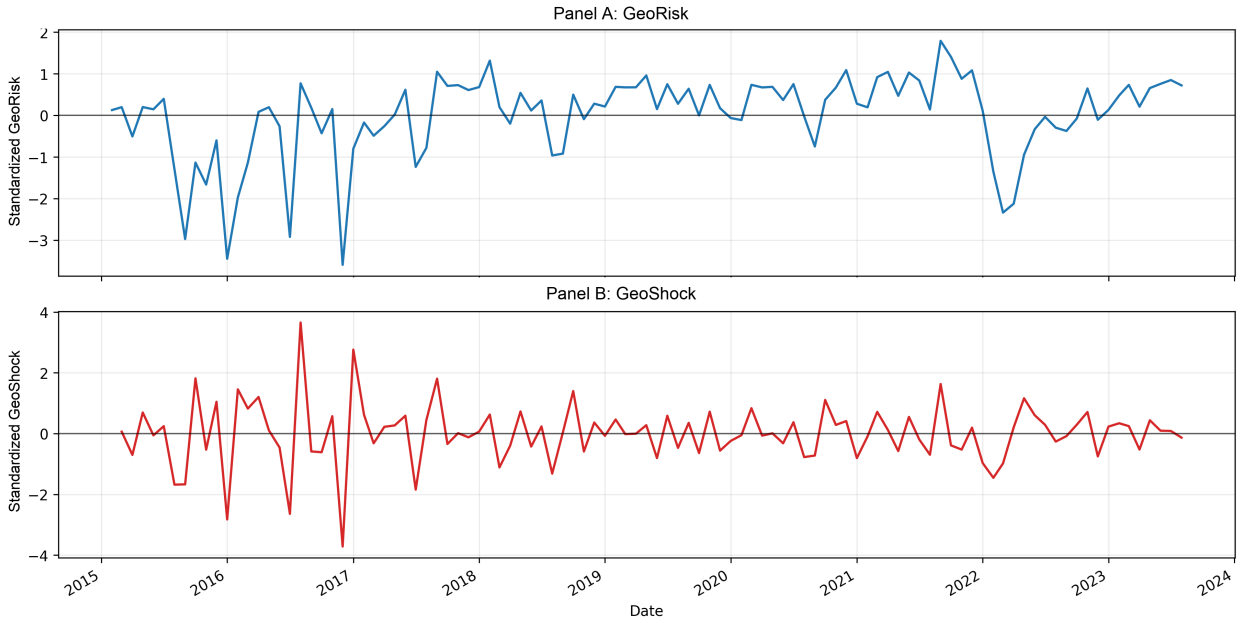


Figure 7: Country-Level GeoRisk and GeoShock Series for Germany

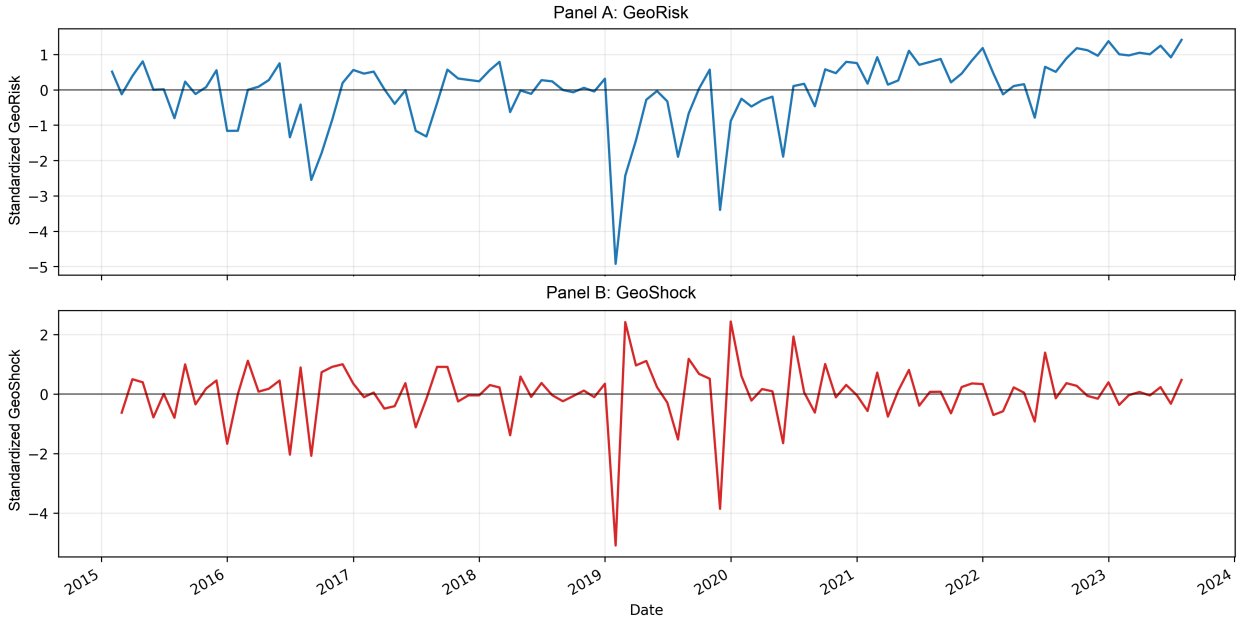


Figure 8: Country-Level GeoRisk and GeoShock Series for India

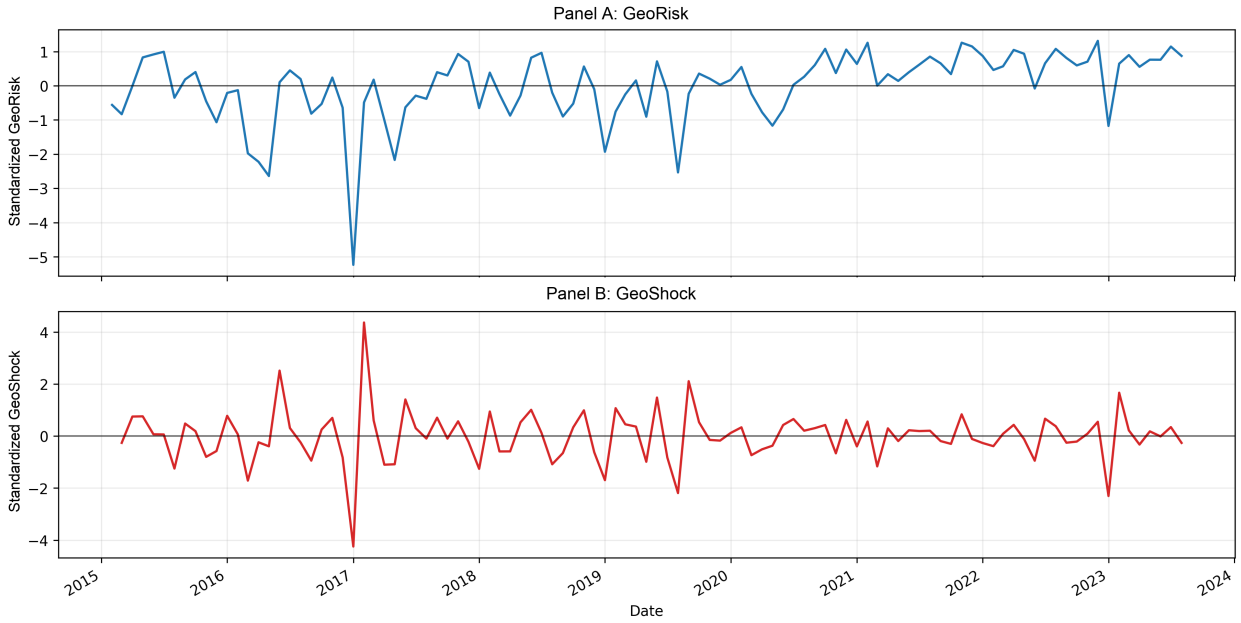


Figure 9: Country-Level GeoRisk and GeoShock Series for Brazil

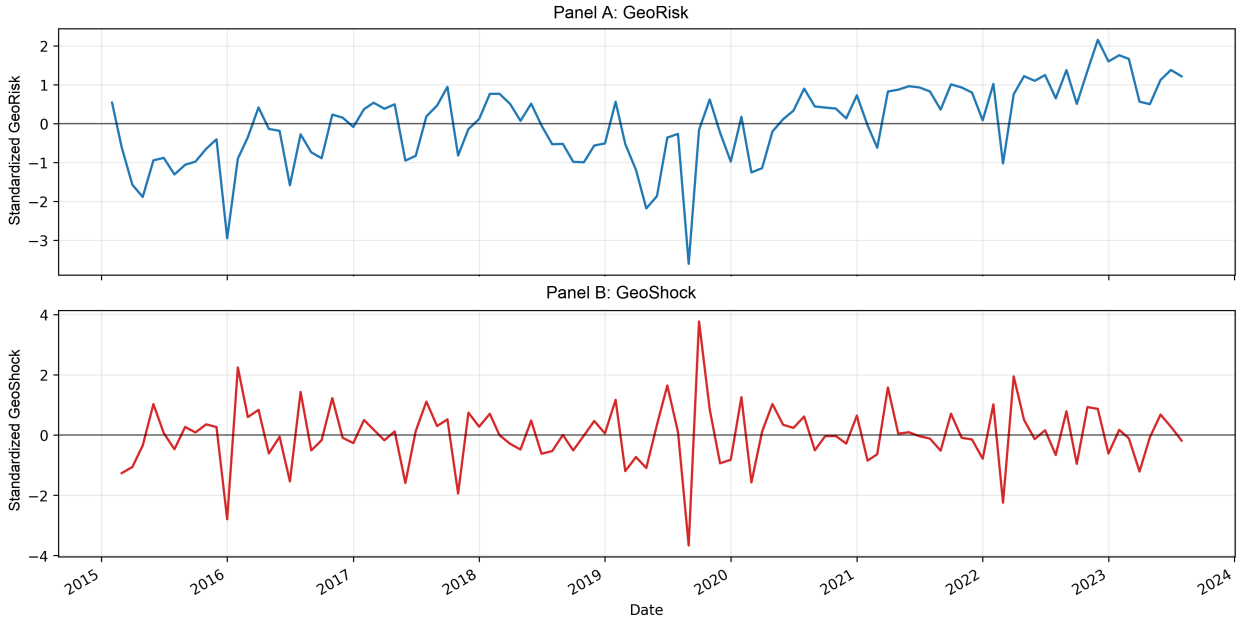


Figure 10: Country-Level GeoRisk and GeoShock Series for Saudi Arabia

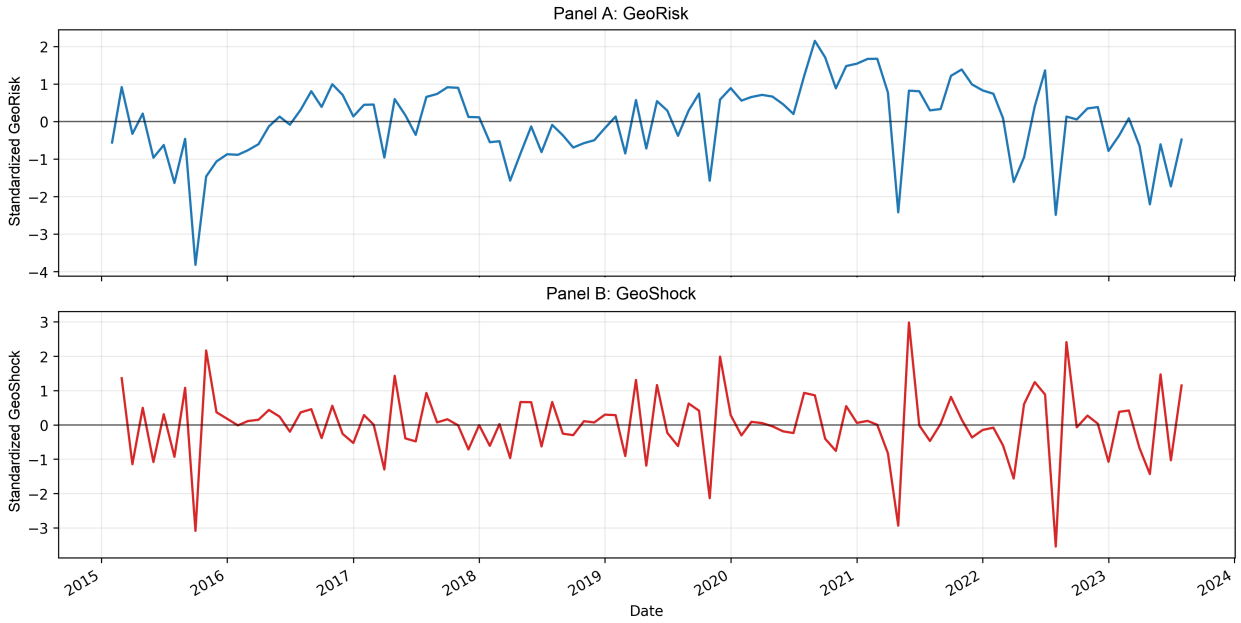


Figure 11: Country-Level GeoRisk and GeoShock Series for Israel

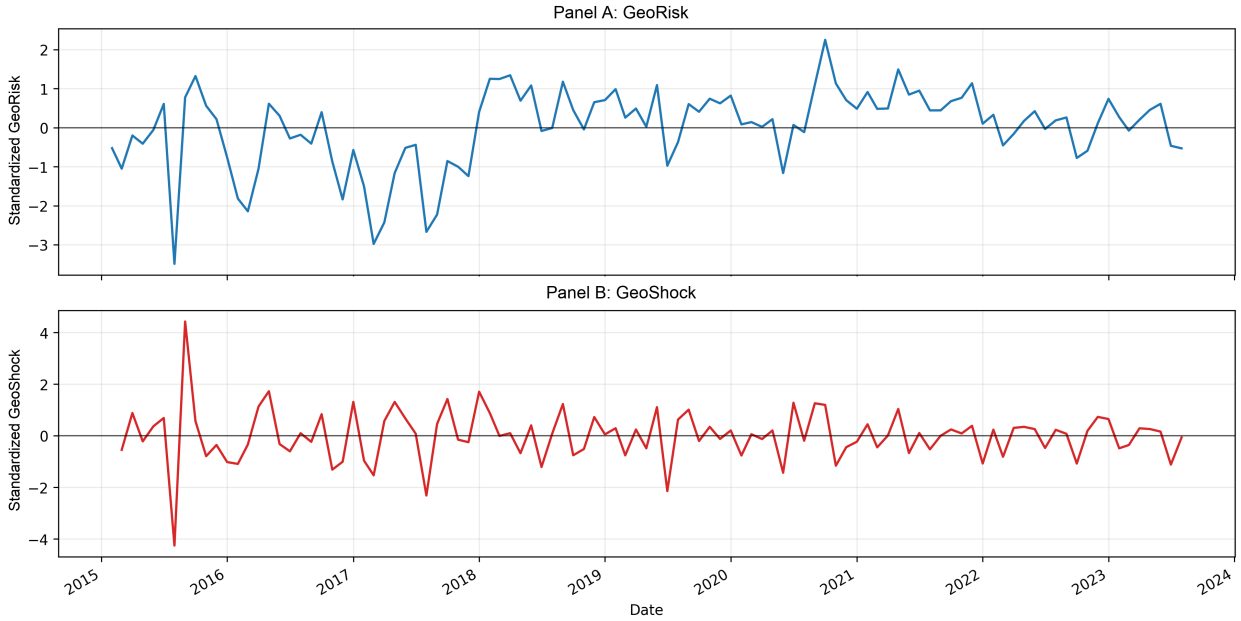


Figure 12: Country-Level GeoRisk and GeoShock Series for South Korea

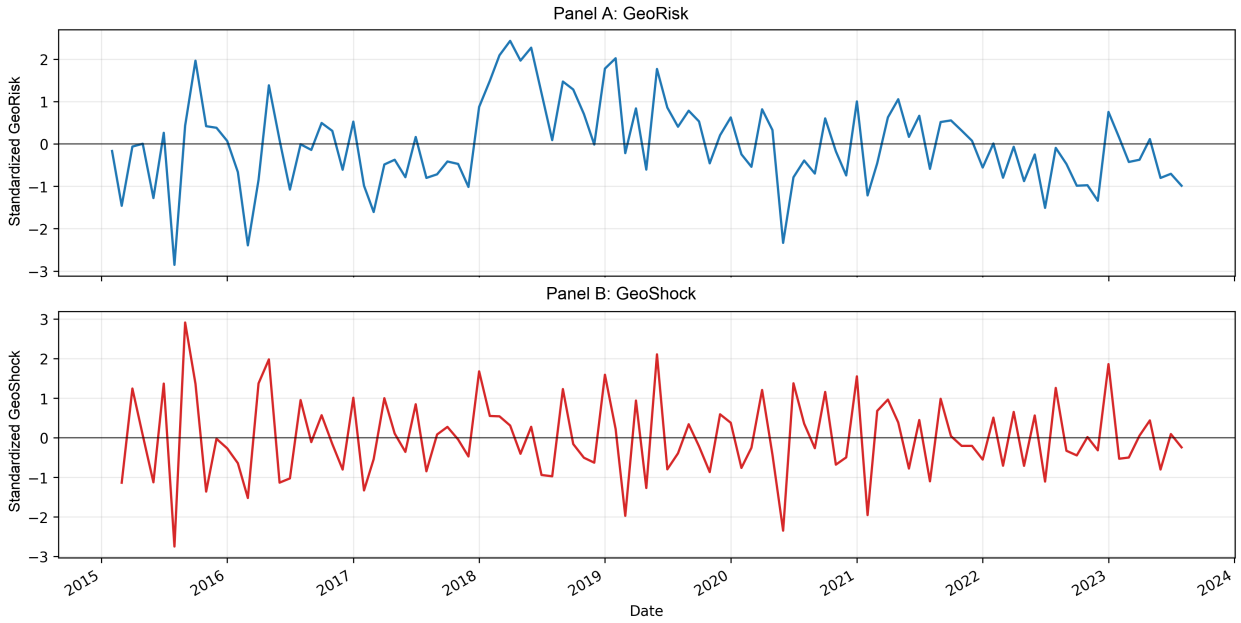


Figure 13: Country-Level GeoRisk and GeoShock Series for North Korea

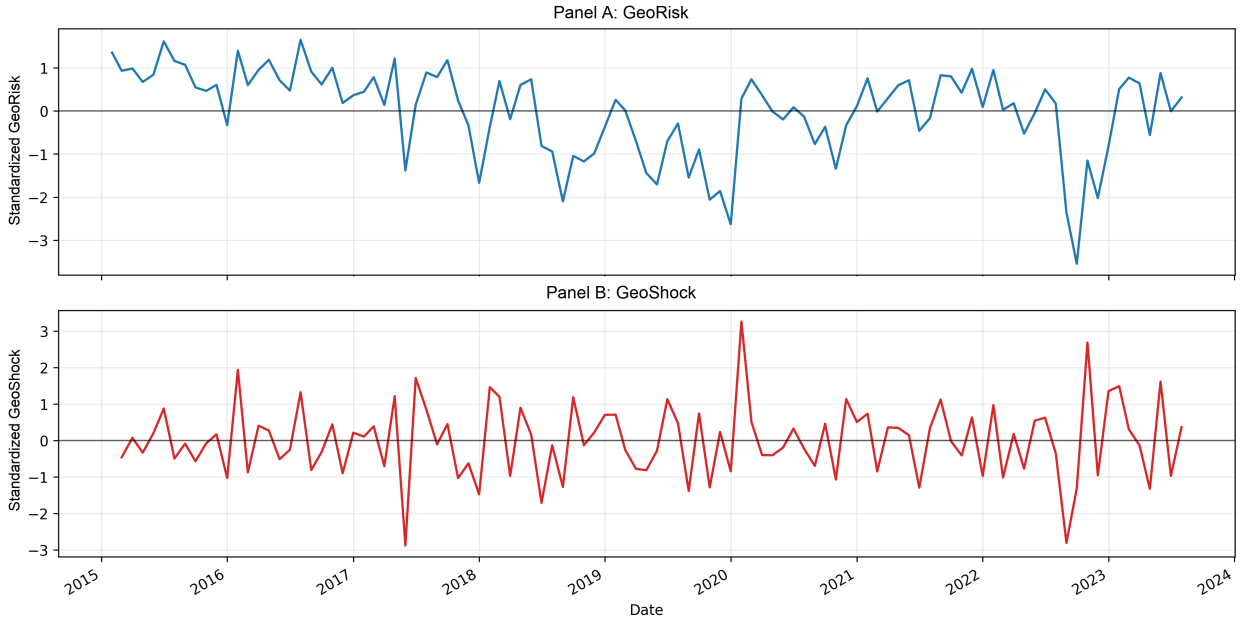


Figure 14: Country-Level GeoRisk and GeoShock Series for Iran

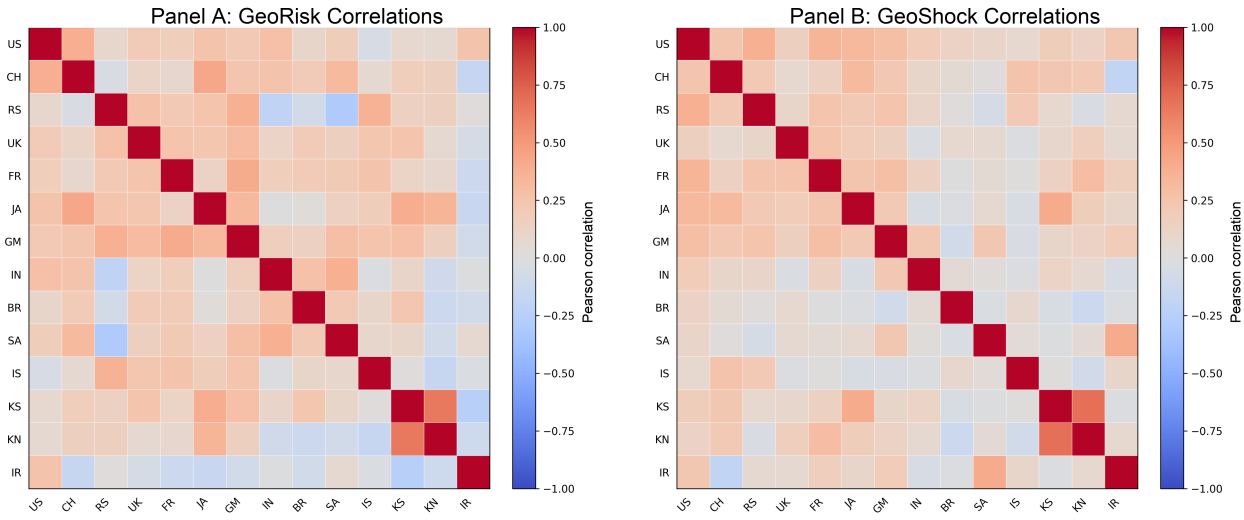


Figure 15: Pairwise Correlations of Country-Level GeoRisk and GeoShock Series. Panel A reports pairwise correlations of standardized monthly GeoRisk series, and Panel B reports pairwise correlations of standardized monthly GeoShock series, over 2015–2023. Darker red indicates higher positive correlation and darker blue indicates higher negative correlation.

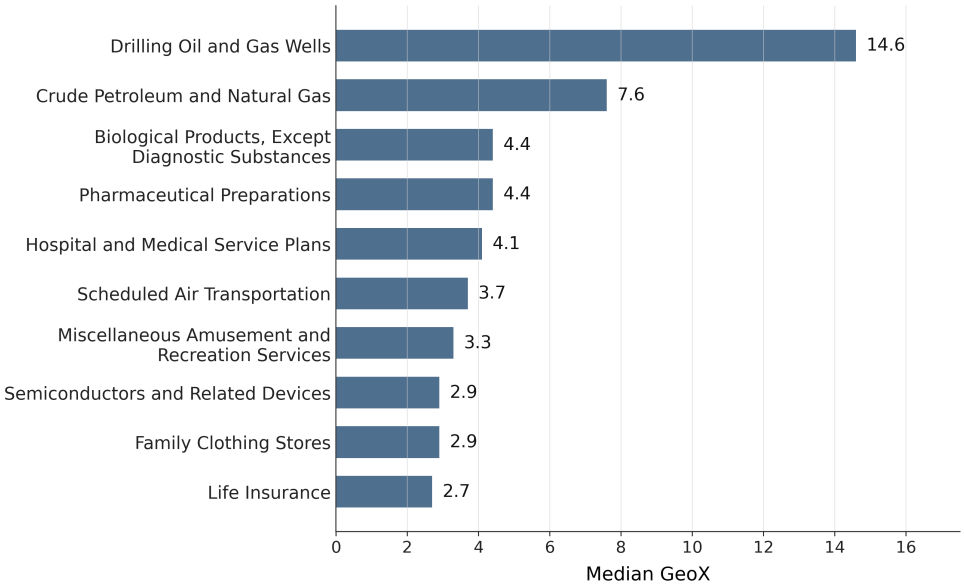


Figure 16: Industry-Level GeoX Rankings. This figure reports the ten SIC industries with the highest median GeoX among industries with at least five sample firms. Bars report medians across firms within each industry. GeoX is the firm-level geopolitical exposure score constructed from statistically significant country-specific geopolitical betas estimated over February 2015–August 2023.

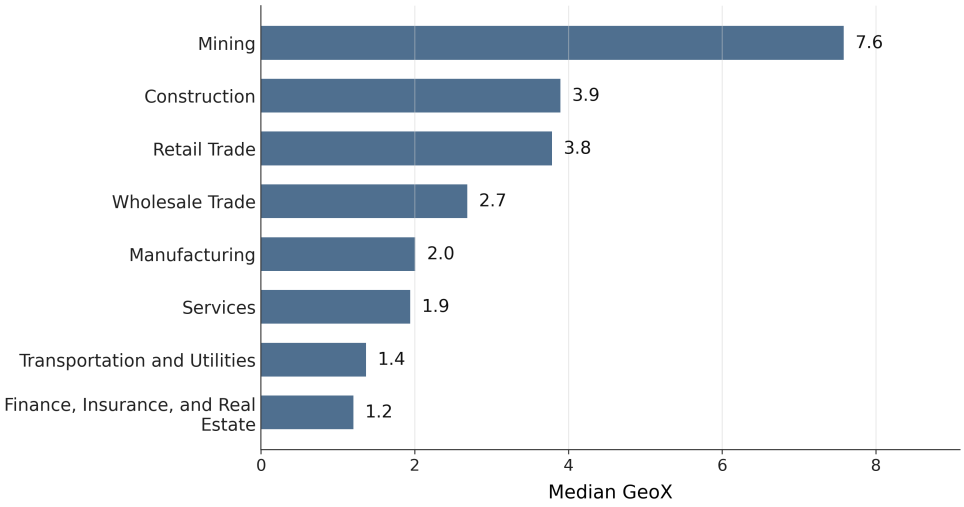


Figure 17: SIC Division-Level GeoX Rankings. This figure reports median GeoX by SIC division among divisions with at least five sample firms. Bars report medians across firms within each division. GeoX is the firm-level geopolitical exposure score constructed from statistically significant country-specific geopolitical betas estimated over February 2015–August 2023.

Table 1: Sample Structure and Descriptive Statistics

Panel A: Sample Structure

Metric	Value
Firm-month observations	61,226
Unique firms	621
Unique months	103
Average months per firm	98.6
Sample period start	Feb. 27, 2015
Sample period end	Aug. 31, 2023

Panel B: Descriptive Statistics

Variable	N	Mean	Std.	P25	Median	P75	Min	Max
Excess return	61,196	1.092	9.856	-4.120	1.035	6.088	-99.844	147.931
Log market cap	61,093	9.906	1.211	9.159	9.809	10.596	4.141	14.926
Log total assets	61,201	9.831	1.414	8.882	9.739	10.687	2.439	15.190
Debt-to-assets	61,201	0.651	0.213	0.504	0.649	0.794	0.154	1.276
ROA	61,196	0.059	0.086	0.017	0.050	0.099	-0.284	0.325
Current ratio	51,043	1.780	1.203	1.031	1.433	2.110	0.385	7.164
Market-to-book	61,093	4.856	12.767	1.656	3.022	5.863	-57.042	74.482

Notes: This table reports the sample structure and descriptive statistics for the S&P 500 constituent sample used to estimate GeoX. The sample includes firms that were constituents at any point during the February 2015–August 2023 estimation period and have at least 48 monthly return observations. Panel A summarizes the firm-month sample. Panel B reports descriptive statistics for monthly excess returns and firm characteristics. Excess returns are expressed in percent. Market capitalization and total assets are measured as natural logarithms of values in millions of U.S. dollars. Debt-to-assets is total debt divided by total assets, ROA is return on assets, current ratio is current assets divided by current liabilities, and market-to-book is market capitalization divided by book equity.

Table 2: Top 25 Firms by Firm-Level GeoX

Rank	Firm	SIC industry	GeoX
1	Noble Corp Plc	Drilling Oil & Gas Wells	40.71
2	Denbury Inc	Crude Petroleum & Natural Gas	40.24
3	Ovintiv Exploration Inc	Crude Petroleum & Natural Gas	39.22
4	Viacom Inc	Television Broadcasting Stations	28.67
5	Concho Resources Inc	Crude Petroleum & Natural Gas	27.29
6	Valaris Limited	Drilling Oil & Gas Wells	27.28
7	Anadarko Petroleum Corp	Crude Petroleum & Natural Gas	27.18
8	Diamond Offshore Drilling	Drilling Oil & Gas Wells	25.06
9	Avon Products	Perfumes, Cosmetics & Toilet Preparations	23.81
10	Garrett Motion Inc	Motor Vehicle Parts & Accessories	23.41
11	Frontier Communications Parent	Telephone Communications	22.58
12	GameStop Corp	Retail–Computer Software Stores	20.95
13	Noble Energy Inc	Crude Petroleum & Natural Gas	20.37
14	Bed Bath & Beyond Inc	Retail–Home Furnishings & Equipment	20.15
15	Transocean Ltd	Drilling Oil & Gas Wells	14.59
16	TechnipFMC Plc	Oil & Gas Field Services, NEC	14.11
17	Dollar Tree Inc	Retail–Variety Stores	13.41
18	Cimarex Energy Co	Crude Petroleum & Natural Gas	13.41
19	Macy's Inc	Retail–Department Stores	13.26
20	EOG Resources Inc	Crude Petroleum & Natural Gas	13.26
21	E-Trade Financial Corp	Security Brokers & Dealers	13.05
22	CNX Resources Corp	Crude Petroleum & Natural Gas	12.80
23	Kohl's Corp	Retail–Department Stores	12.67
24	Mallinckrodt Plc	Pharmaceutical Preparations	12.65
25	Genworth Financial Inc	Life Insurance	12.28

Notes: This table reports the 25 firms with the highest GeoX scores in the historical S&P 500 constituent sample. GeoX is the firm-level geopolitical exposure score constructed from statistically significant country-specific geopolitical betas estimated from monthly excess-return regressions over February 2015–August 2023. Because GeoX aggregates absolute betas, higher values indicate stronger historical return sensitivity to country-level geopolitical shocks, regardless of whether the estimated sensitivities are positive or negative. The historical constituent sample includes both entrants and exits during the estimation period.

Table 3: Out-of-Sample Return Regression with GeoX–GeoShock Interactions

	Coefficient	Std. Error	t-stat	p-value
Constant	-4.486**	1.551	-2.892	0.004
GeoX	-0.163*	0.094	-1.729	0.084
GeoX × GeoShock (US)	-0.245	0.304	-0.808	0.419
GeoX × GeoShock (China)	0.362*	0.209	1.734	0.083
GeoX × GeoShock (Russia)	-0.017	0.123	-0.135	0.892
GeoX × GeoShock (UK)	-0.526	0.411	-1.278	0.201
GeoX × GeoShock (France)	-0.262	0.264	-0.990	0.322
GeoX × GeoShock (Japan)	0.248	0.192	1.293	0.196
GeoX × GeoShock (Germany)	0.755	0.464	1.626	0.104
GeoX × GeoShock (India)	-0.108	0.170	-0.634	0.526
GeoX × GeoShock (Brazil)	0.151	0.188	0.804	0.421
GeoX × GeoShock (Saudi Arabia)	-0.903**	0.422	-2.139	0.032
GeoX × GeoShock (Israel)	0.453**	0.203	2.230	0.026
GeoX × GeoShock (South Korea)	0.090	0.153	0.587	0.557
GeoX × GeoShock (Iran)	0.313	0.236	1.325	0.185
GeoX × GeoShock (North Korea)	-0.129	0.213	-0.607	0.544
Month FE	Yes			
Observations	12,467			
R^2	0.029			
Adj. R^2	0.026			

Notes: This table reports pooled OLS estimates for September 2023–June 2025. The dependent variable is monthly firm excess return, expressed in percent. GeoX is the firm-level geopolitical exposure score estimated over February 2015–August 2023 and held fixed during the out-of-sample period. GeoShock denotes the standardized monthly country-level GeoShock series for the indicated country. Interaction terms estimate whether the relation between country-level GeoShock series and firm excess returns varies with GeoX. Month fixed effects absorb shocks common to all firms in a month, including the standalone GeoShock series. Standard errors are conventional. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

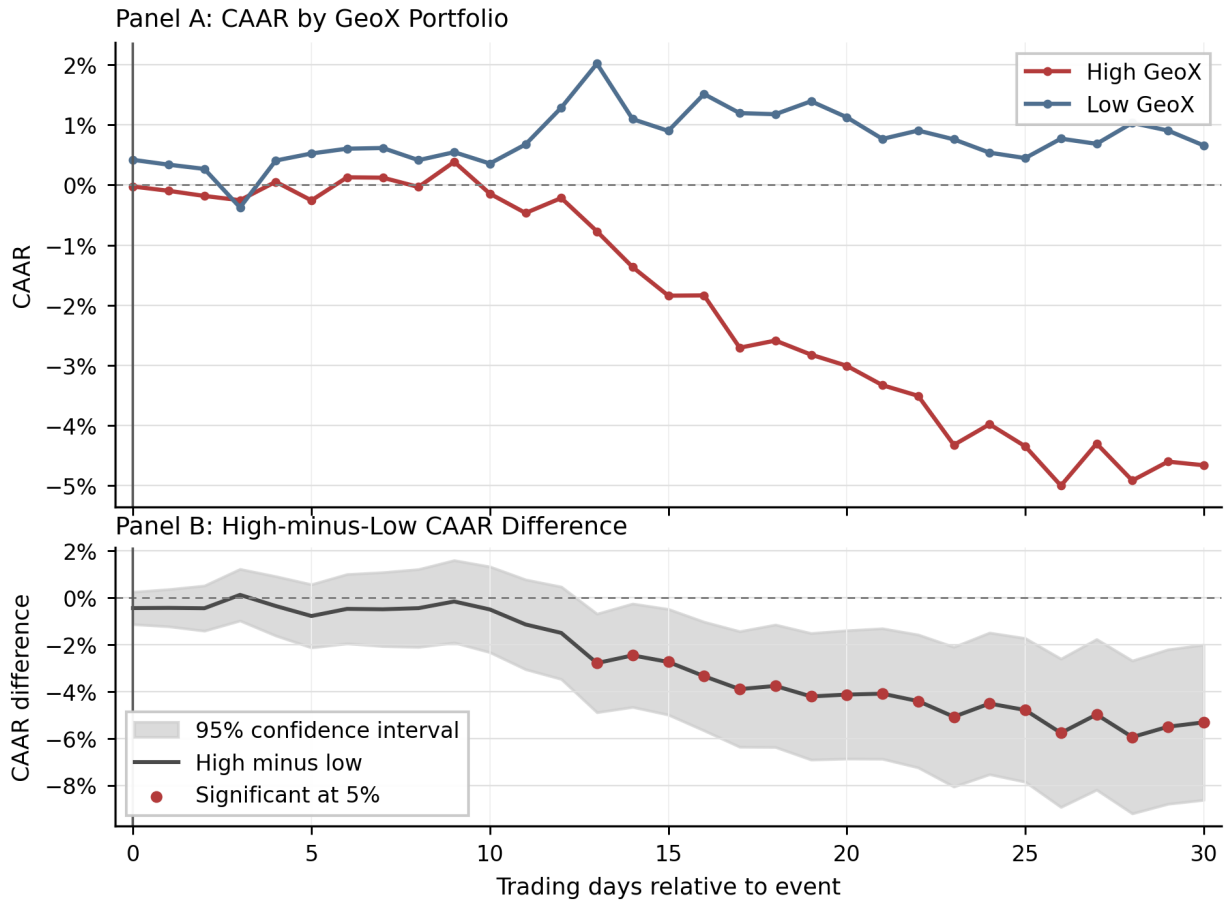


Figure 18: Event-Study CAARs Around the October 7, 2023 Middle East Shock. Panel A plots cumulative average abnormal returns for firms in the top and bottom quartiles of nonzero GeoX over the 30 trading days following the event. Panel B plots the high-minus-low CAAR difference with 95% confidence intervals. High-GeoX firms experience persistent losses relative to low-GeoX firms, and the high-minus-low CAAR difference remains negative through the event window.

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A Appendix: Geopolitical Themes

This appendix lists the set of geopolitical themes used to construct the country-level GeoRisk and GeoShock series. Each theme reflects a distinct category of geopolitical activity or discourse captured from GDELT Global Knowledge Graph 2.0. The descriptions summarize the types of events or issues covered under each theme.

Geopolitical Theme	**Description**
ARMEDCONFLICT	Primary discussion of armed conflict, from ground assaults to air raids
BLOCKADE	Mentions of blockades and no fly zones; use in combination with the military themes to distinguish armed vs non-kinetic blockades
CEASEFIRE	Mentions of ceasefires, truces, peace deals, suspension of hostilities, etc
CYBER_ATTACK	Any discussion of cyberwarfare, cyberattacks, phishing, hacking, hacktivists, viruses, etc
ECON_FOREIGNINVEST	Foreign investment
ECON_FREETRADE	Free trade
ECON_NATIONALIZE	Discussion of nationalization of industry, etc
ECON_TRADE_DISPUTE	Trade disputes, antidumping, etc
ELECTION	Elections of any kind
ELECTION_FRAUD	Election fraud, from ballot stuffing to vote rigging
ENV_OIL	Oil power

Geopolitical Theme	**Description**
EXTREMISM	Discussion of extremists, both political (far right/far left/anarchist), groups like skinheads, and terms like zealot or "fringe element"
FREESPEECH	Freedom of speech
IMPEACHMENT	Impeachment
INTERNET_BLACKOUT	Mentions of internet blackouts
INTERNET_CENSORSHIP	Internet censorship such as blocking websites, etc
MEDIA_CENSORSHIP	Censorship of the mainstream media, such as arresting journalists, press freedoms, etc
MILITARY	General discussion of the military
MILITARY_COOPERATION	Actual mentions of military ties, military aid, defense cooperation, etc
NEGOTIATIONS	Discussion of bargaining, negotiations, etc
PEACEKEEPING	Mentions of peacekeepers, UN forces, etc
PERSECUTION	Actual mentions of "persecution"
POLITICAL_PRISONER	Explicit mention of political prisoners or labor camps
POLITICAL_TURMOIL	Discussion of major unrest such as martial law, political "crisis," state of emergency, violent suppression, etc
PROPAGANDA	Explicit mention of "propaganda"
PROTEST	Discussion of protesting, demonstrating, rioting, striking, activists, agitators, etc
REBELLION	Mutinies, uprisings, civil war, etc

Geopolitical Theme	**Description**
REFUGEES	Refugees, displaced persons, forced migration, asylum seekers
RELEASE_HOSTAGE	A hostage being released
RETALIATE	Actual mentions of "reprisals," "avenge," "strike back," etc
SANCTIONS	Sanctions and embargoes
SIEGE	Sieges, besieging, blockades, checkpoints, etc
SOC_DIPLOMCOOP	Diplomatic cooperation
SOC_ECONCOOP	Economic cooperation
SOC_INTELSHARING	Intelligence sharing
SOVEREIGNTY	Actual mentions of "sovereignty," "proclaim independence," "independent state," etc
STATE_OF_EMERGENCY	Martial law, state of emergency, emergency rule, etc
SUICIDE_ATTACK	Suicide attacks
SURVEILLANCE	Surveillance, spying
TAX_TERROR_GROUP	Returns any mention of a major recognized terror group around the world
TERROR	Primary discussion of terrorists, bombings, assassinations, etc
TREASON	Treason, sedition, insurrection
VETO	Actual mentions of "veto," "voted against," etc
VIOLENT_UNREST	Discussion of violent unrest, from rubber bullets, firing into a crowd, clashes with police or soldiers, violent rioting, etc
WATER_SECURITY	Drought, water access, lack of rain, clean water, etc

Geopolitical Theme	**Description**
WMD	Weapons of mass destruction, from nuclear to biological to chemical warfare